GAME DESIGN & LEVEL SPECIFICATION DOCUMENT (GDD)

Still Life

Surrealist Selective-Empathy & Zero-Slack Counterweight Puzzle Labyrinth

Author / Role	Kevin (Lefi) Shan — Solo Game & Level Designer, Programmer	**Engine & Stack**	Unity 6 (URP), C#, Custom HLSL Kuwahara/Ink Shaders
Production Scope	Solo Prototype · USC CTIN (Situated Role: The Dreamer)	**Facility Scope**	200m Linear-Z Facility (5 Rooms + 3 Hallways) · 235 kg Zero-Slack Math



Figure 1.1: Room II Three-Story Menagerie Archive where living book flocks flutter past rigid animal vaults.

Executive Summary & Design Intent

Inside a monochromatic gallery-foundry, biological roles are inverted: animals (5 kg Chicken to 100 kg Horse) stand frozen as mute, taxidermy-still counterweights, chasm planks, and gear wedges, while a 1.4 kg Desk Lamp shivers in the dark, flees in terror, and leans into your hand for warmth.

Rather than lecturing the player about empathy through dialogue, the level design manufactures a behavioral blind spot: 30 minutes of weighing, bridging, and wedging animals trains the player to treat animals strictly as kilograms—until Room IV's shredder rejects a 25 kg Dog as a 'Living Organism' and leaves you holding the only inanimate object in the room.

1. Core Design Pillars & The L.O.P. Gameplay Loop

Section 1.1 – 1.3

1.1 World as Enigma (The Dreamer)

In Peter Brinson's taxonomy, the Dreamer inhabits a world governed by dream logic—an enigmatic internal coherence that players act within before grasping its ethical weight. Inspired by René Magritte's object paradoxes and Inside, surreal contradictions are framed as standard archival procedure.

1.2 Selective Empathy Inversion

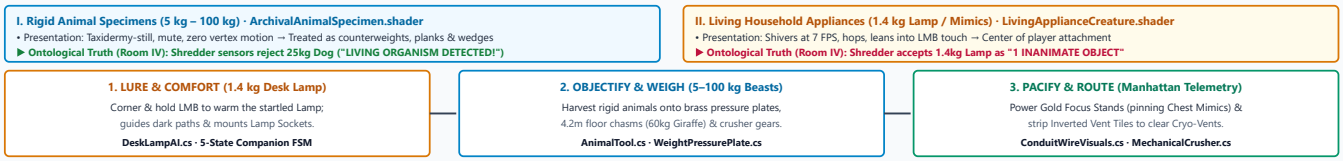
Animals have zero vertex animation, zero sound, and strict 5 kg mass increments, inviting purely instrumental use. Conversely, the 1.4 kg Desk Lamp shivers at 7 FPS, hops with a 1.24m gait, and warms up under LMB rubbing—monopolizing the player's protective instincts.

1.3 The L.O.P. Core Loop

(1) Lure & Comfort the 1.4 kg Lamp to light dark paths and power Lamp Sockets; (2) Objectify & Weigh rigid animals (5–100 kg) onto brass plates, across 4.2m chasms, and into crusher gears; (3) Pacify Chest Mimics & Cryo-Vents via Manhattan conduits so the Lamp can follow.

Figure 1.2 — System Architecture: Selective Empathy Inversion & The 3-Step L.O.P. Core Loop

Surface Presentation vs. Room IV Sensor Ground Truth



2. Specimen & Appliance Counterweight Specification Catalog

Table 2.1 — Strict 5 kg Multiples vs. 1.4 kg Anomaly

Catalog ID	Specimen / Appliance	Mass	Mechanical Classification	Systemic Function Across Rooms I–V
SPEC-01	**Chicken**	5 kg	Calibration Shim	Fine-grain 5 kg increment for 15 kg / 55 kg composite locks in Rooms II & III.
SPEC-02	**Cat**	10 kg	Light Counterweight	Powers Hallway I Focus Stand (10 kg) and pairs with Chicken (5 kg) for 15 kg locks.
SPEC-03	**Penguin**	15 kg	Standard Counterbalance	Single-unit 15 kg activator for 2F West Dog Vault, 1F SE Elevator, and Gold Focus Plates.
SPEC-04	**Dog**	25 kg	Doorstop & Finale Anchor	Opens 25 kg gates in Rooms II–III; rejected by Room IV shredder before awakening in Room V.
SPEC-05	**Deer**	40 kg	Phase 3 Vault Key	Unlocked in Room II Back-West Dual Vault; powers the 40 kg 3F Penthouse Gold Focus Plate.
FIG. 02	**Giraffe (Blueprint #2)**	60 kg	4.2m Bridge & Heavy Weight	Tips horizontally across 2F & 3F 4.2m chasms as a walkable plank OR weighs 60 kg on plates.
FIG. 03	**Tiger (Blueprint #3)**	80 kg	Lift Ballast & Gear Wedge	Powers Room II 80 kg 2F→3F Elevator and jams Room III Central Mechanical Crusher gears.
FIG. 01	**Horse (Blueprint #1)**	100 kg	Path Blocker & Max Ballast	Rises from Room I floor hatch to corner the fleeing Lamp and locks Room III Upper Flywheel.
FIG. 04	**Desk Lamp (Blueprint #4)**	1.4 kg	**Companion & "Only Object"**	**Shivers in cold/dark and hops behind player, yet sensors classify it as an inanimate object.**

3. Master Facility Architecture (Z = -12m to +188m) & 5-State Desk Lamp FSM

5 Rooms & 3 Induction Hallways

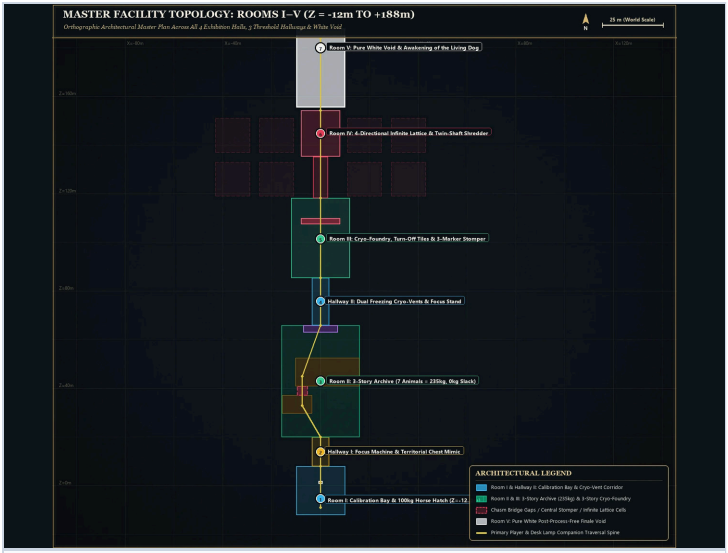


Figure 3.1 — Master Facility Plan (Z = -12m to +188m): Orthographic layout across Room I (Study), Hallway I, Room II (3-Story Archive), Hallway II, Room III (Cryo-Foundry), Hallway III, Room IV (Infinite Lattice), and Room V (Pure White Void).

3.1 Linear-Z Threshold Progression

The 200m facility alternates between monumental multi-story puzzle chambers (Rooms I–V) and narrow single-mechanic induction corridors (Hallways I–III):

- **Room I & Hallway I (Z = -12m..+38m):** 100kg Horse hatch ambush, LMB Lamp comforting, and 10kg Cat Focus Stand pinning a Chest Mimic.
- **Room II (Z = +38m..+66m):** Three-story vertical archive (28m × 28m × 13.8m) housing 7 animals (235 kg total) across 4 zero-slack phases.
- **Hallway II & Room III (Z = +66m..+118m):** Freezing Cryo-Vents, Inverted "Turn-Off" Vent Tiles, and the 3-Marker Colossal Central Stomper.
- **Hallway III, Room IV & V (Z = +118m..+188m):** 4-directional infinite lattice, Industrial Shredder moral reversal, and Pure White Living Dog awakening.

3.2 Room I Onboarding Without UI Popups

Approaching the Desk Lamp triggers a 0.68s neck crane and 0.52s defensive flashbang as it leaps off the desk and flees. Rather than letting players chase it forever, a 15s timer drops open the central floor hatch and raises the 100 kg Plaster Horse—physically cornering the shivering Lamp so the player discovers LMB drag-comforting.

Figure 3.2 — Desk Lamp 5-State Companion FSM & Environmental Fear Interrupts (DeskLampAI.cs)

Procedural Hopping Gait (1.24m Step), LMB Drag-Warming & Mimic/Cryo Fear Stops



3.3 Technical Art, Manhattan Conduit Telemetry & Mesh Consolidation

Table 3.1 — Shaders & Rendering Systems

Shader / Technical System	Target Domain	Implementation & Design Function
`ArchivalAnimalSpecimen.shader`	Rigid Animals (5–100 kg)	Triplanar procedural ink cross-hatching + deadpan plaster diffuse ramp + brass pedestal ring (zero vertex motion).
`LivingApplianceCreature.shader`	Living Appliances (Lamp, Mimics)	Quantized 7 FPS hand-drawn vertex boil + cold shivering displacement + warm subsurface blush from _StillLife_LampPosWS.
`ConduitDirectionalPulse.shader`	Zero-HUD Manhattan Telemetry	Orthogonal floor/wall conduits color-coded by circuit: Cyan (Gates/Lifts), Gold (Focus Stands), Green (Stomper Jammers).
`Room4AvantGardeInfiniteBuilder`	4-Way Infinite Megastructure	Consolidates hundreds of 4-directional Hallway III & Room IV replica cells into 6 UInt32 static meshes (~6 draw calls, 200+ FPS).



Figure 3.3 — Room I (100 kg Horse Hatch): Rising from the floor hatch at t = 15s to corner the fleeing Desk Lamp.

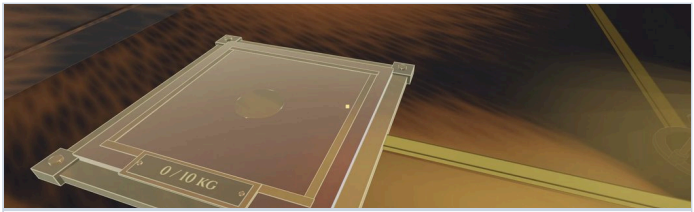


Figure 3.4 — Hallway I (Focus Stand & Mimic): 10 kg Cat on the Gold Plate powers the Focus Stand, levitating the Chest Mimic.

4.1 Room II: The Three-Story Archive — 4-Phase 235 kg Zero-Slack Combinatorics

`1F Courtyard, 2F Mezzanine & 3F Penthouse (0 kg Idle Slack)`

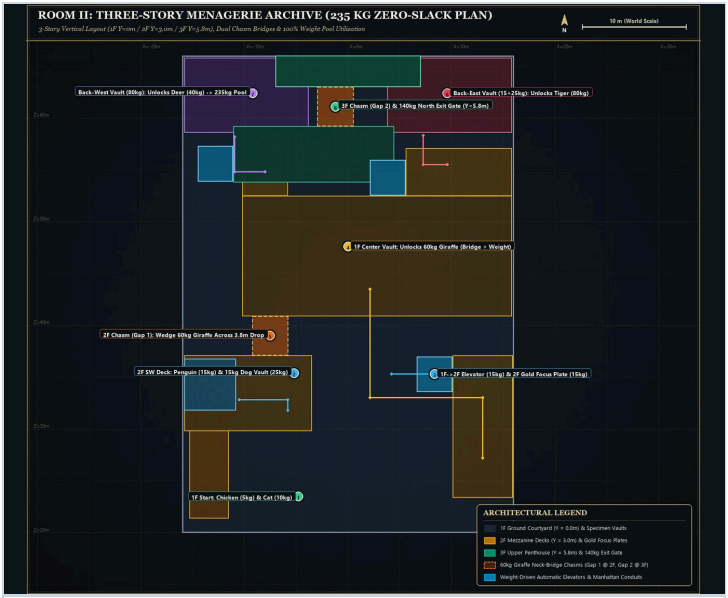


Figure 4.1 — Room II Level Design Map (1F, 2F & 3F): Showing all 7 animal vaults (235 kg total), 3 Gold Focus Stands, 2 vertical elevators, and the 2F/3F 4.2m Giraffe chasm bridges.

Design Problem: Eliminating Surplus-Mass "Physics Cheese"

In multi-prop weight puzzles, unlocking heavier weights usually creates surplus mass that trivializes later doors. To guarantee every specimen remains essential, Room II enforces a strict Zero-Slack Invariant: at the climax of each phase (55 kg → 115 kg → 195 kg → 235 kg), 100% of your currently unlocked animal pool is simultaneously locked across elevators, focus stands, chasm bridges, and vault gates (0 kg idle mass).

Dual-Role Specimen: The 60 kg Giraffe Bridge & Counterweight

Unlocked at the end of Phase 1, the elongated 60 kg Giraffe can be tipped horizontally across the 4.2m floor chasms on 2F and 3F as a physical plank, OR reclaimed into your inventory to weigh 60 kg on guarded vault plates. After unlocking the 40 kg Deer in Phase 3, the Giraffe wedges permanently across the 3F chasm to prevent upper-floor soft-locks.

Mathematical Proof: Why Phase 3 (Deer) Cannot Be Skipped

Reaching the 3F Exit requires keeping the 2F→3F Elevator powered (-80 kg Tiger), bridging the 3F chasm (-60 kg Giraffe), pinning the 3F Chest Mimic (-40 kg Gold Plate), and opening the 3F Exit Gate (-55 kg). Without unlocking the 40 kg Deer in Phase 3, the player is 40 kg short (195 kg < 235 kg).

4.1.2 Four-Phase 100% Weight Budget Allocation Table (55 kg → 115 kg → 195 kg → 235 kg)

`Table 4.1 — Simultaneous Lock Equations`

Progression Phase	Unlocked Animal Pool	Simultaneous Active Locks (100% Allocation Equation)	Idle Slack & Reward
Phase 1: Pre-Giraffe	`55 kg` 5 Chick + 10 Cat + 15 Peng + 25 Dog	`-15 kg` 1F Gold Focus Plate (5+10) + `-15 kg` 1F SE Lift (15 Peng) + `-25 kg` 2F SE Release Gate (25 Dog) = **`55 / 55 kg`**	**0 kg Slack** Unlocks: 60 kg Giraffe
Phase 2: Pre-Tiger	`115 kg` 55 Smalls + 60 Giraffe	Bridge 2F chasm w/ Giraffe → Flip 2F SwitchBox → Reclaim Giraffe: `-15 kg` 2F Gold Plate + `-60 kg` 2F NE Outer Gate (Giraffe) + `-40 kg` 1F NE Inner Gate (25+15) = **`115 / 115 kg`**	**0 kg Slack** Unlocks: 80 kg Tiger
Phase 3: Pre-Deer	`195 kg` 55 Smalls + 60 Gir + 80 Tig	`-80 kg` 2F→3F West Lift (80 Tiger) + `-60 kg` 3F Overlook Outer Gate (60 Giraffe) + `-55 kg` 1F NW Inner Gate (25+15+10+5) = **`195 / 195 kg`**	**0 kg Slack** Unlocks: 40 kg Deer
Phase 4: 3F Finale	**`235 kg`** All 7 Animals Active	`-80 kg` 2F→3F Lift (Tiger) + `-60 kg` 3F Chasm Bridge (Giraffe Wedged) + `-40 kg` 3F Gold Plate (Deer) + `-55 kg` 3F Exit Gate (25+15+10+5) = **`235 / 235 kg`**	**0 kg Slack** **Room II Complete!**

4.1.3 In-Game Execution: Atrium Overview, Focus Beam Traversal & 2F Giraffe Bridge

`Figures 4.2 – 4.4`



Figure 4.2 — 3-Story Archive Atrium: Fluttering book flocks and vertical vault tiers.



Figure 4.3 — Carrying 60 kg Giraffe: Passing the 1F Gold Focus Stand beam.



Figure 4.4 — 4.2m Giraffe Bridge: Tipping the 60 kg Giraffe across the 2F gap.

4.2 Hallway II & Room III: The Cryo-Foundry, Inverted Vent Tiles & 3-Marker Stomper

`225 kg Foundry Pool & Mechanical Jamming`

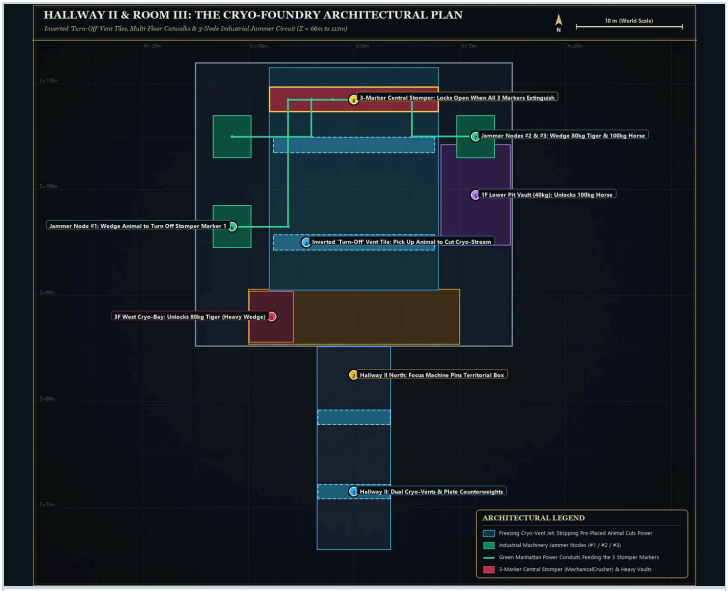


Figure 4.5 — Hallway II & Room III Level Design Map: Showing Freezing Cryo-Vents, Inverted "Turn-Off" Vent Tiles, and the 3 green Manhattan conduits feeding the Colossal Central Stomper.

Hallway II Induction: Asymmetric Cryo-Vent Hazard

Hallway II introduces Freezing Air Vents (FreezingAirVent.cs) that blast icy streams across the path. While the human player can walk straight through cold jets unharmed, the 1.4 kg Desk Lamp shivers violently and freezes at the edge of the stream—requiring the player to depress a shutoff plate with the 25 kg Dog so the Lamp can cross.

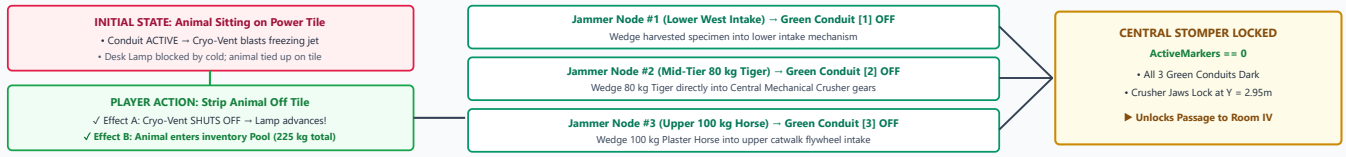
Subverting Plate Logic: Inverted "Turn-Off" Vent Tiles

In Rooms I–II, placing weight ON a plate always unlocked a path. Room III reverses this habit: pre-placed animal specimens sit on Inverted Power Tiles (ControlledByPowerTile) that actively power the room's freezing jets. Picking an animal UP off its tile simultaneously shuts off the cryo-vent for the Lamp AND harvests that animal into your counterweight pool.

3-Marker Colossal Central Stomper & Mechanical Jamming

At the heart of the foundry, a Colossal Central Stomper (MechanicalCrusher.cs) slams shut until all 3 green Manhattan conduit markers are extinguished by jamming three sub-mechanisms: (1) Lower West Jammer Node, (2) Mid-Tier Crusher Gears (wedged by the 80 kg Tiger), and (3) Upper Flywheel Intake (locked by the 100 kg Horse).

Figure 4.6 — Room III Systemic Circuit: Inverted Vent Harvesting → 3-Node Mechanical Jamming Unifying Hazard Deactivation & Counterweight Harvesting Into One Loop



4.2.3 In-Game Execution: Cryo-Vents, Central Stomper, 80 kg Tiger Jam & 100 kg Horse Lock

`Figures 4.7 – 4.10`



Figure 4.7 — Hallway II Cryo-Vent: 25 kg Dog on plate cuts the freezing jet.



Figure 4.8 — Central Stomper: Guarded by 3 active green conduit markers.



Figure 4.9 — 80 kg Tiger Jam: Wedging the Tiger into Jammer Node #2.



Figure 4.10 — 100 kg Horse Lock: All 3 green conduits extinguished.

4.3 Room IV & V: The Industrial Shredder Moral Reversal & Pure White Awakening

`Subverting 30 Minutes of Player Conditioning`

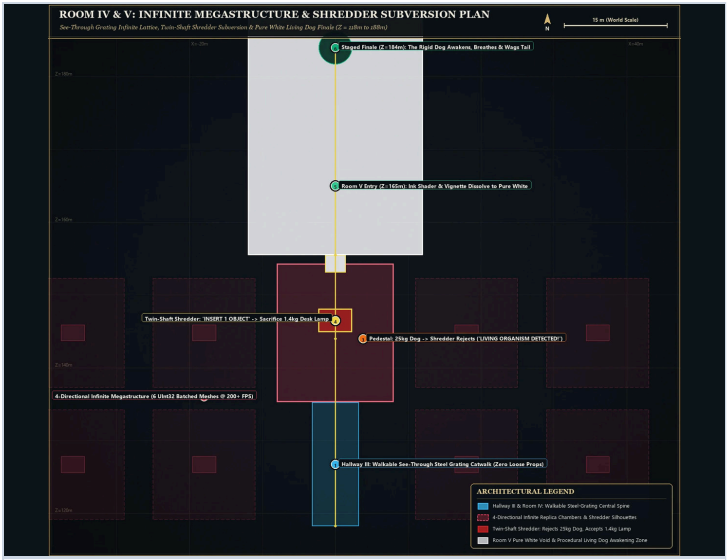


Figure 4.11 — Hallway III, Room IV & Room V Level Design Map: 4-Directional Infinite Megastructure (Z = +118m..+156m), the Industrial Shredder, and Room V Pure White Void (Z = +156m..+188m).

Weaponizing Player Conditioning: The Shredder Trap

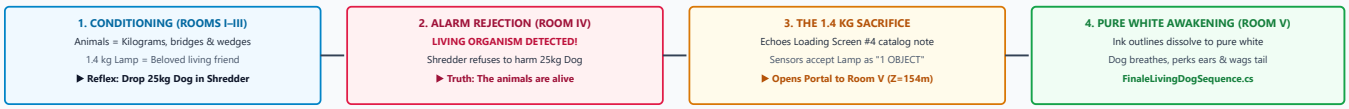
Before entering Room IV, Loading Screen #4 displays FIG. 04 — DESK LAMP (1.4 KG): "It shivers in the dark and leans into your hand like something alive, but to the gallery's sensors it is still just an object." Inside Room IV's 4-directional infinite steel lattice, an Industrial Shredder blocks the exit with the prompt: INSERT 1 OBJECT TO PROCEED. Beside it sits the 25 kg Dog.

Trained for 30 minutes to treat animals as expendable hardware, players instinctively drop the 25 kg Dog into the hopper—only for the shredder to blare "BZZZT! SAFETY MECHANISM ACTIVATED: LIVING ORGANISM DETECTED!" and eject the Dog unharmed. With zero loose props in the room, the truth lands without a single dialogue box: the mute animals were alive all along, and your shivering 1.4 kg Desk Lamp is the only inanimate object the sensors will accept.

Room V: Pure White Dissolve & Living Dog Awakening

Sacrificing the 1.4 kg Desk Lamp opens the portal at Z = +154m. Stepping into Room V smoothly lerps PureWhiteFinaleSuppression (1.0 → 0.0), dissolving the oppressive Kuwahara ink outlines into a pure white void where the 25 kg Dog awakens—breathing, perking its ears, tilting its head, and wagging its tail.

Figure 4.12 — The 4-Stage Psychological Conditioning & Unscripted Reversal Arc (Rooms I–V) How 30 Minutes of Mechanical Habit Set Up a Single Unscripted Realization



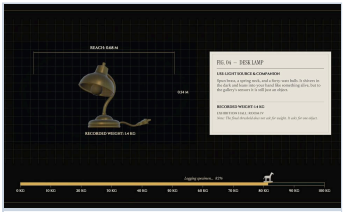


Figure 4.13 — FIG. 04 Blueprint: "To the sensors it is just an object."



Figure 4.14 — Dog Rejected: "LIVING ORGANISM DETECTED!"

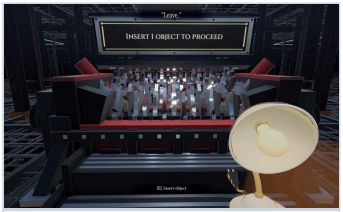


Figure 4.15 — 1.4 kg Sacrifice: Holding the shivering Lamp at the chute.



Figure 4.16 — Room V Awakening: The Living Dog breathes & wags its tail.

5. Design Retrospective & Key Takeaways

`Post-Mortem Synthesis`

5.1 Conditioning Over Exposition

Room IV's twist works because the level design never lectures the player. Thirty minutes of weighing, bridging, and wedging animals manufactures the player's blind spot—so when the shredder rejects the 25 kg Dog as a living organism, the player catches their own reflex in the act.

5.2 Zero-Slack Math Prevents Cheese

Enforcing 0 kg idle slack across all four phases of Room II (55 → 115 → 195 → 235 kg) guaranteed that unlocking heavier specimens never made lighter ones obsolete; every animal had to be continuously re-routed across all three floors.

5.3 Zero-HUD Manhattan Telemetry

Color-coding orthogonal floor and wall conduits by circuit function—Cyan (gates/lifts), Gold (Focus Stands), and Green (Stomper jammers)—made complex multi-floor dependencies readable at a glance without floating UI markers.